MIL-G-25728(USAF)
13 February 1957

MILITARY SPECIFICATION

GAGE, PRESSURE, DIAL INDICATING, OXYGEN, 0-500 PSI, GENERAL SPECIFICATION

1. SCOPE

1.1 This specification covers the performance requirements and general construction of oxygen system pressure gages in the range 0-500 pounds per square inch.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

BB-O-925	Oxygen, 99.5 Perdent, Gas Liquid
NN-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment)
LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid (For Domestic Shipment)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-676	Boxes, Set-Up, Paperboard

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
JAN-P-103	Packaging and Packing for Overseas Shipment - Boxes; Wood Cleated, Solid Fiberboard
JAN-P-108	Boxes, Fiberboard, Corrugated and Solid (Military Type)
MIL-P-116	Preservation, Methods Of
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant
MIL-B-138	Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)

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MIL-B-4229	Boxes; Paperboard, Metal-Stayed
MIL-D-5028	Drawings and Data Lists; Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-N-6011	Nitrogen; Liquid and Gas
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads; Standard, Aeronautical
MIL-T-9107	Test Reports; Preparation Of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of

Air Force-Navy Aeronautical

AND10053	Bosses; Pipe-Thread, Internal-Standard Dimensions For
AND10398	Metals-Definition of Dissimilar

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Selection of Specifications and Standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143 except as provided in 3.2.1 and 3.2.2.

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3.2.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard Parts.- With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials

3.3.1 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.1.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in AND10398.

3.3.2 Nonferrous.- Nonferrous materials shall be used for all parts of the gage, except where ferrous materials are essential and do not come in direct contact with the oxygen.

3.3.3 Nonmagnetic.- Nonmagnetic materials shall be used for all parts of the instruments except where magnetic materials are essential.

3.4 Design.- The gage shall be so designed that no parts will work loose in service. They shall be leakproof, and shall be built to withstand, without failure, the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Pivots, bearings, and gears shall neither bind nor shake, and shall be as nearly frictionless as practicable. The gages shall be so designed that minor repairs, such as replacement of pointer, can be easily made by the personnel of the operating units without

3.5 Construction

3.5.1 Threads

3.5.1.1 Screw Threads.- Screw threads shall conform to MIL-S-7742.

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3.5.1.2 Pipe Threads.- Pipe threads shall conform to MIL-P-7105.

3.5.2 Mechanism Adjustment.- The mechanism shall be provided with a means to adjust or correct the indications in case this becomes necessary during the service of the gage. Adjustment shall require no special tools.

3.5.3 Pointer Adjustment.- The pointer shall be light and sufficiently rigid to prevent oscillation under vibration, and shall be free to swing beyond either end of the range without being engaged by a mechanical stop. Suitable stops shall be incorporated within the mechanism to prevent the sector from being disengaged from the pinion. The pointer shall be firmly attached to the mechanism, yet shall be readily adjustable. The pointer shall be of such length as to touch but never overlap the graduations on the dial face. The indicating end of the pointer shall be painted white.

3.5.4 Overpressure Stop.- An overpressure stop shall be provided on the gage, designed in such a manner as to restrain the mechanism rather than the pointer. The overpressure stop shall allow pointer travel beyond full scale to 525, plus or minus 25, pounds per square inch. When extreme overpressure occurs, the stop shall not allow the pointer to move beyond the bottom center of the dial.

3.5.5 Hair Spring.- The hair spring shall be made from phosphor bronze.

3.5.6 Surge Pressure Restriction.- A restriction of 0.020 inch diameter shall be provided between the gage connection and gage mechanism.

3.5.7 Dial.- The dial shall be fastened securely upon the frame of the mechanism by at least two screws, in such a manner that it will not loosen or turn when the gage is vibrated.

3.5.7.1 Dial Face.- The dial face shall be as shown in the applicable detail specification.

3.6 Performance

3.6.1 Scale Error At Room Temperature.- When tested as specified in 4.5.2, the gage shall be capable of indicating the oxygen pressure in cylinders of low-pressure oxygen systems of aircraft at room temperature. The gage shall read within the scale tolerance specified in section 4. Tolerances shall not exceed those specified either before or after tapping the gage.

3.6.2 Friction.- The gage shall be tested for friction as specified in 4.5.3.

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3.6.3 Low Temperature.- The gage shall function properly at a temperature of minus 65 degrees Fahrenheit, when tested as specified in 4.5.5.

3.6.3.1 Vibration.- When tested as specified in 4.5.5 the gage shall be capable of operating, while being vibrated in a linear or circular motion at a double amplitude of 0.018 to 0.020 inch, at a frequency varying from 300 to 3,000 cycles per minute while being maintained at a temperature of minus 65 degrees Fahrenheit, plus or minus 10 degrees Fahrenheit.

3.6.4 Cold Resistance and Low Temperature.- The pointer shall move smoothly through the range without damage to the gage while being exposed to a temperature of minus 85 degrees Fahrenheit, when tested as specified in 4.5.6.

3.6.5 High Temperature.- When tested as specified in 4.5.7, the gage shall function properly at a temperature of plus 160 degrees Fahrenheit.

3.6.6 Seasoning.- The gage shall function satisfactorily after being subjected to the seasoning tests as specified in 4.5.8.

3.6.7 Overpressure.- The gage shall function satisfactorily after being subjected to the overpressure tests specified in 4.5.9.

3.6.8 Leakage.- The gage shall not leak upon completion of all the above performance tests, when tested as specified in 4.5.4.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with each other with respect to installation and performance. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.8 Weight.- The maximum weight of the gage shall be as specified in the applicable detail specification.

3.9 Finish.- Protective coating and finishes which will crack, chip, or scale during normal service life, or owing to extremes of atmospheric conditions, shall not be used.

3.10 Markings.- All markings shall be sufficiently durable to withstand usage encountered in service.

3.11 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.12 Workmanship.- Workmanship shall be in accordance with high-grade commercial aircraft instrument manufacturing practice. The gage shall be thoroughly cleaned of all oils and foreign matter, and shall be free from any defect that might affect appearance of functionability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of gages shall be classified as follows:

a. Acceptance tests ..... See 4.2
b. Preproduction testing ..... See 4.3

4.2 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.2.1 Individual Tests.- Each gage shall be subjected to the following tests as described under Test Methods of this specification.

a. Examination of product
b. Scale error at room temperature
c. Friction
d. Leakage

4.2.2.1 Lot.- A lot shall consist of 200 gages, or fraction thereof, manufactured under essentially the same conditions and submitted for acceptance at substantially the same time.

4.2.2.2 Sampling Plan.- Two gages shall be selected at random from each lot and subjected to the tests listed below and described under Test Methods.

a. Low temperature and vibration
b. Cold resistance and low temperature
c. High temperature
d. Seasoning
e. Overpressure

4.2.2.2.1 Rejection and Retest.- When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary test shall be repeated.

4.2.2.2.2 Individual Tests May Continue.- For production reasons individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

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4.2.3 Defects in Items Already Accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 Preproduction Testing

4.3.1 Preproduction Test Sample Tested by the Contractor.- The contractor shall subject two gages to the preproduction tests specified in 4.3.4.

4.3.2 Preproduction Test Report.- After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.3.3 Preproduction Test Sample for the Procuring Activity.- Along with the preproduction test report, the contractor shall submit two samples to the procuring activity.

4.3.4 Preproduction Tests.- Preproduction tests shall consist of all tests described under Test Methods.

4.4 Test Conditions

4.4.1 Atmospheric Conditions.- Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees Fahrenheit, plus or minus 18 degrees Fahrenheit, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for change in instrument reading.

4.4.2 Gas.- The gas used in the testing of the gages shall be either oxygen conforming to type I, grade A, of BB-O-925; water pumped nitrogen conforming to MIL-N-6011; or water pumped air equivalent in dryness to MIL-N-6011 nitrogen. If nitrogen or air is used, appropriate density correction factors shall be applied to the flowmeter to make allowance not only for the effect on the meter but also on the performance of the gage with the lower density gas.

4.4.3 Tapping.- Unless otherwise specified, the gage shall be tested in position and shall be lightly tapped before a test reading is taken.

4.4.4 Position.- All tests shall be conducted in the horizontal position unless stated otherwise.

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4.4.5 Vibration.- Whenever vibration is specified, it shall be at any specified frequency between 300 and 3,000 cycles per minute in a circular motion; and at a double amplitude of not less than 0.018 nor more than 0.020 inch.

4.5 Test Methods

4.5.1 Examination of Product.- Unless otherwise specified by the procuring activity, each gage shall be examined to determine conformance with the requirements of this specification with respect to general construction, material, workmanship, weight, finish, and markings.

4.5.2 Scale Error At Room Temperature.- The gage shall be tested for scale errors at the points on the scale indicated by an asterisk in table I, and shall not exceed the corresponding tolerances. Tests shall be made by subjecting the gage to the pressures required to produce these readings, first with pressures increasing, then with pressures decreasing. With the pressures increasing, the pressure shall be brought up to, but shall not exceed the pressure required to give the desired readings. With the pressure decreasing, the pressure shall be brought down to, but shall not fall below, the pressure required to give the desired readings. At the discretion of the Government inspector, gages may be tested for scale error at any additional point or points on the scale.

TABLE I

SCALE ERROR AT ROOM TEMPERATURE

TEST PRESSURE	TOLERANCE PLUS OR MINUS (PSI)
0*	10
20	10
40	10
60*	10
80*	10
100*	10
200*	25
300*	25
400*	25
500*	25

* Test Points

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4.5.3 Friction.- The gage shall be tested for friction at all points indicated by an asterisk in table I, beginning with the second asterisk point. The specified tolerances shall not be exceeded. The pressure shall be increased in order to bring the pointer approximately to the desired reading and then held constant while two readings are taken; the first before the gage is tapped, and the second after the gage is tapped. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the gage. This test may be combined with the test for scale error at room temperature. Failure to meet this test will be cause for rejection.

4.5.4 Leakage.- Each gage shall be subject to a pressure of at least 500 pounds per square inch for a period of 5 minutes. This pressure shall be locked in the gage and the supply disconnected as close to the gage as practicable. The hand shall then be observed for deviation from the original pressure setting. Any drop of pressure than can be attributed to leaks in the gage will be cause for rejection.

4.5.5 Low Temperature and Vibration.- The gages shall be subjected to a temperature of minus 65 degrees Fahrenheit and subjected to vibration for a period of 3 hours on a suitably-designed vibration stand. The vibration shall be in any plane with respect to the instrument with a double amplitude of from 0.018 to 0.020 inch and frequency of 300 to 3,000 cycles per minute. After the 3 hour period of vibration, and while the gages are still being vibrated at the specified temperature, they shall be tested for scale error as described in the scale error at room temperature test. The readings shall not differ from the individual test readings by more than plus or minus 10 pounds per square inch from 0 to 100 pounds, or plus or minus 25 pounds per square inch from 200 to 500 pounds.

4.5.6 Cold Resistance and Low Temperature.- The gages shall be placed in a chamber maintained at a temperature of minus 85 degrees Fahrenheit for a period of 48 hours. At the end of this period, and while still at that temperature, the gages shall be tested for operation to determine that the pointers move smoothly throughout the entire range of travel and to determine that no component parts have been damaged. The temperature shall then be raised to minus 65 degrees Fahrenheit for a period of not less than 24 hours, after which time and while at this temperature, the gage shall be tested for scale error and friction using the methods specified in 4.5.2 and 4.5.3. The readings obtained shall not differ from the individual test readings by more than plus or minus 20 pounds per square inch at 0 to 100 pounds and plus or minus 50 pounds per square inch from 200 to 500 pounds. Following the return to room temperature, the gages shall be subjected to the individual tests.

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4.5.7 High Temperature.- The gages shall be subjected to a temperature of plus 160 degrees Fahrenheit for a period of 24 hours. The gages shall then be subjected to the methods of the scale error at room temperature test while at high temperature, and the readings observed shall not exceed the individual test readings by more than plus or minus 10 pounds per square inch from 0 to 100 pounds, or plus or minus 25 pounds per square inch from 200 to 500 pounds. Following return to room temperature, the gages shall be subjected to the individual tests. The gages shall then be examined, and any evidence of deterioration or damage caused by the high temperature exposure, such as melting of solder, blistering, or streaking of paint, etc., shall be cause for rejection.

4.5.8 Seasoning.- The gage shall be subjected to 300 applications of pressure from 0 to 500 pounds per square inch. Not less than 1 hour later, the gage shall be tested for scale error and friction as specified in 4.5.2 and 4.5.3. The readings obtained after the applications of the differential pressure shall not differ from those obtained in the individual tests by more than 10 pounds per square inch.

4.5.9 Overpressure.- The gage shall be subjected to a pressure of at least 700 pounds per square inch for a period of 10 minutes. Not less than 1 hour after the release of the overpressure, that gage shall be subjected to the individual tests specified in 4.2.1. The readings obtained in the scale error at room temperature and friction tests shall not differ from those obtained in the original individual tests by more than plus or minus 10 pounds per square inch from 0 to 100 pounds or plus or minus 25 pounds per square inch from 200 to 500 pounds.

5. PREPARATION FOR DELIVERY

5.1 Levels of Preservation and Packaging

5.1.1 Level A.- Gages shall be preserved in accordance with MIL-P-116, method II, and packed in containers meeting MIL-B-4229, PPP-B-566, PPP-B-676, or JAN-P-108. For domestic shipment, containers meeting LLL-B-631 or LLL-B-636 may be used. Unit quantities shall be as specified by the procuring activity.

5.1.2 Level C.- Gages shall be preserved and packaged in accordance with manufacturer's commercial practice.

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5.2 Intermediate Packaging.- Only identical items shall be included in an intermediate package. For items preserved and packaged to meet 5.1.2, requirements for intermediate packaging shall be at the option of the contractor. Gages preserved and packaged to meet 5.1.1 shall be packaged in containers meeting LLL-B-631, LLL-B-636, PPP-B-676, MIL-B-4229, JAN-P-108, or PPP-B-566. Unless otherwise specified by the procuring activity, the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the limitations of the container being used.

5.3 Levels of Packing

5.3.1 Level A.- Gages preserved and packaged to meet 5.1.1 and 5.2 shall be packed in exterior type shipping containers meeting JAN-P-103, PPP-B-621, JAN-P-108, MIL-B-138, PPP-B-585, PPP-B-601, or MIL-B-10377. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to JAN-P-108 and is sealed at all joints and seams, including manufacturers' joint, with tape conforming to JAN-P-127.

5.3.2 Level B.- Gages preserved and packaged to meet 5.1.1 and 5.2 shall be packed in domestic type exterior containers meeting NN-B-591, PPP-B-601, PPP-B-621, PPP-B-585, LLL-B-631, LLL-B-636, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shapes and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall be the special requirements table of LLL-B-631 or LLL-B-636, as applicable.

5.3.3 Level C.- Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules, or regulations of other common carriers as applicable to the mode of transportation.

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5.4 Physical Protection.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100 except that for domestic shipment, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of JAN-P-100 shall be waived when preservation, packaging and packing of the item are for immediate use, or when drop tests of MIL-P-116 are applicable.

5.5 Marking.- Exterior and interior packages shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use.- The pressure gages covered by this specification are intended for use in indicating the oxygen supply pressure in low pressure oxygen systems in aircraft.

6.2 This specification covers general construction and performance requirements only. The dial, case (when applicable), weight, and dimensional requirements, and other special requirements, are covered in the applicable detail specifications.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer, use or sell any patented invention that may in any way be related thereto.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-25728(AS)	GAGE,PRESSURE,DIAL INDICATING,OXYGEN,0-500 PSI

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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